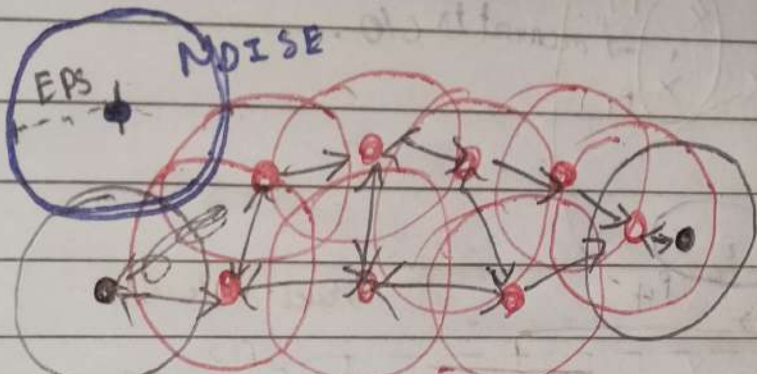
DBSCAN.

Density-Based Spatial Clustering of Applications with Noise



• ϵ (epsilon)

↳ neighbourhood radius

• Minpts

↳ min num. of pts required to form a dense region

• Min Pts = 4

→ For assigning minpt, first create a Boundary around a point & give r = Epsilon (eps).

• Core Point → If its ϵ- neighbour

→ It lies in dense region & can can expand a cluster.

• Border Point → It is not a core pt.

Red: Core Points

Yellow → Border Pts, still part of a cluster because, its → within epsilon of core point, but not does meet minpts criteria

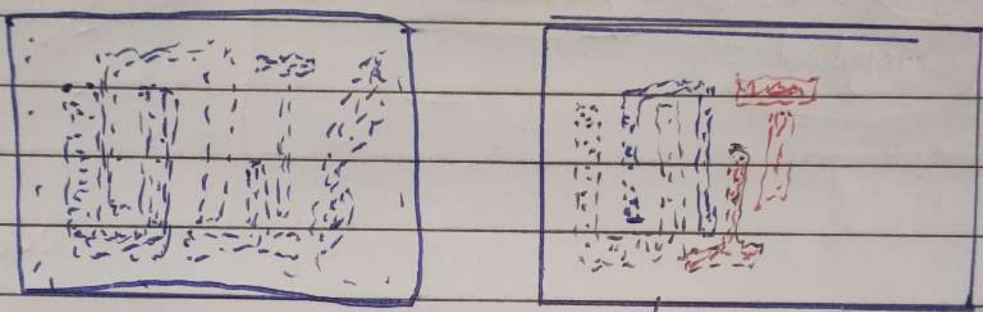
Blue: Noise point, not assigned to a cluster

· Advantages of DBSCAN.

→ Is great at separating cluster of high density versus clusters of low density within a given d-set.

→ Great with Handling Outliers

↳ ∵ multi dimentionalities can be shown using DBSCAN, say



↓ more visible dimensions

· Disadvantages of DBSCAN.

→ Doesn't work well while working with clusters of diff-dimensions.

· Great at separating high density clusters from low den-clusters

→ Struggles with cluster of same density

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